# Research question:

## How does altering the pH (3-7) using citric acid affect the germination of *Vigna radiata*, measured by the mass (g) after five days?

# Biology Internal assessment

**Word Count: 2937**

## Introduction

Acid Rain

As the need for electricity rapidly grew in the past few decades, the increasing burning of coal to match the energy demand resulted in power plants being the major contributor to the release of sulfur dioxide ($SO_2$) and nitrogen oxides ($NO_2$) (United States Environmental Protection Agency). $SO_2$ and $NO_2$ react with water in the atmosphere to form sulfuric and nitric acids, which directly causes a pressing environmental issue, acid rain. Its impact is far-reaching, affecting aquatic ecosystems, soil chemistry, infrastructure, and plant health by changing the optimal pH in these systems. Plants as the keystone species of many terrestrial ecosystems are severely impacted by the changing pH of soil and water, leading to forest decline and crop yield reduction(Shi et al.).This not only disrupts the food chain for natural ecosystems, but also impact humans as it threatens agriculture and jeopardizes our food supply. Therefore, my research aims to investigate how changing pH impacts the growth of plants by using citric acid to mimic the effects of acid rain on mung bean germination. Better understanding of this helps develop strategies to mitigate acid rain's effects, and support ecosystem restoration efforts to ensure food security and environmental stability.

Germination

Seed germination is the first and crucial step of plant growth that is strongly related to seedling survival rate. Through the process of water uptake known as imbibition which triggers various metabolic processes, a seed shifts away from dormant and begins to grow actively(Ali and Elozeiri). Mung beans *(Vigna radiata*) are particularly suitable for studying germination due to their rapid growth and widespread use in agriculture and nutrition. The optimal environment for mung bean growth is in a pH range of 6.3-7.2 and at temperature between 28°–35°C (Bhardwaj et al.)

Effect of low pH

Acidic environment, simulated by changing the concentration of citric acid($C_6H_8O_7$) disrupts the germination of seeds due to its ability to intervene enzyme activity. Enzymes called amylase play the crucial role of enabling the respiration process of which seeds obtain energy to germinate, specifically by catalyzing the hydrolysis process of starch into glucose. However, α-Amylase in mung beans only works well at pH 5.7 and is sensitive to change in abiotic stressors(Tripathi et al.). Hyperacidic environments could denature the enzyme which ultimately hinders the germination process.

## Research question:

How does altering the pH (3-7) using citric acid affect the germination of *Vigna radiata*, measured by the mass (g) after five days?

<u>Hypothesis:</u>
If the pH of the germination environment is decreased using citric acid, then the biomass of germinated *Vigna radiata* after five days will also decrease, because acidic conditions interfere with enzyme activity essential for metabolic processes, reducing energy availability for growth.

## Methodology:

<u>Selection of the growing method:</u>
Hydroponic growing is selected because it's easy to execute and monitor. More importantly, it allows a better control over the environmental factor by elinaming the potential variability introduced by substances in soil.

**Table 1. Independent and Dependent variables**

| | Description | Scope/quantity | Manipulation/ measurement |
|---|---|---|---|
| Independent variable | **pH of the solution:**<br>• The pH is used to measure the acidity or alkalinity of a solution, indicated by the concentration of hydrogen ions ($H^+$) in the solution (United States Environmental Protection Agency, "What Is pH?"). The pH of solution, serving as the water in the germination process, is the independent variable being manipulated to simulate varying acidic conditions caused by acid rain. | • The pH range is 3–7. This range of mildly to strongly acidic conditions is able to cover a realistic spectrum of acid rain, which is typically around 4.0-5.5 (United States Environmental Protection Agency. The interval of 1 is sufficient for a trend to be identified while remaining accurate and practical given the precision of the available pH probe. | • measured using a calibrated pH probe which can measure up to two decimal places.<br>• Citric acid is chosen: safe to handle; it is a weak organic acid that can change the pH effectively without introducing other complex chemical interactions that might occur with stronger acids.<br>• Since pH adjustments are nonlinear(Tang and Karim, vol. 29), the solid citric acid is first dissolved in water to create solution, which allows finer control in pH adjustment. |

| | | | |
|---|---|---|---|
| Dependent variable | **Biomass of *Vigna radiata* seeds (grams):**<br><br>● allows quantitative measurements to indicate the growth and overall germination success. | ● Two seeds are treated as a single sample because a single seed is too light to be measured accurately with the precision of the scale.<br>● 5 days is the typical germination period (Fong et al.)<br>● 12 samples per pH: ensure sufficient data for analysis even with defective seeds | ● Measured using an electronic balance with precision up to 0.01 g.<br>● After 5 days of germination, seeds are dried at 80°C for 40 minutes to allow moisture to be removed. |

**Table 2. Controlled variables**

| **Controlled Variables** | **Reason for control** | **Method of Control** |
|---|---|---|
| Exposure to light | Although light does not directly impact the germination process of mung beans, exposure to intense light may increase the temperature and cause dehydration which directly impacts the germination. | Plates are wrapped in paper towels and placed in a dark, enclosed area to eliminate light exposure. |
| Temperature | Temperature has a great effect on enzymatic activity. Enzymes enable the germination process by facilitating the breakdown of starch into glucose which serves as the energy source. Thus, enzymatic activity impacts the success of germination. | Well plates are kept in the same environment with relatively consistent temperature. Thermometer is used to monitor the change in temperature. Since all seeds experience the same temperature fluctuation, it will not significantly affect the final result. |
| Type of seed | Different species or varieties of seeds can have different germination rates, conditions and growth patterns, directly impacting the measurement of germination. | All beans come from the same packet of the same brand |
| Size of seed | Larger seeds may have more stored energy for sprouting and initial growth compared to smaller seeds. This will result in a greater increase in biomass. | Seeds are carefully chosen to ensure relatively similar size (±1 mm) |
| Source of water | Different sources of water may contain varying substances, such as minerals | All water used is from the same bottle of distilled water |

| | | |
|---|---|---|
| | and have different pH, which impacts the germination | |
| Volume of water | The amount of water received by the seeds influence the germination | The amount of water used is measured with pipette and graduated cylinder. |
| Source of citric acid | Purity of citric acid impacts the germination as impure citric acid introduces other substances and affects pH manipulation | All citric acid are laboratory grade and is obtained from the same container |
| Duration | Germination time directly influences biomass as beans grow as time progresses | All seeds are left to germinate for the same amount of time (5 days) before measurement. |
| Size of containers | The volume of the containers can impact the germination because factors such as space availability, oxygen supply, and moisture retention could vary. | All mung beans are grown in identical well plates with identical size. |

**Table 3. Materials**

| Equipment/Material | Quantity | Size/Uncertainty |
|---|---|---|
| Citric Acid ($C_6H_8O_7$) (solid) | 5 grams | - |
| Distilled Water | 500 ml | - |
| pH Probe | 1 | ±0.01 pH |
| Beakers | 5 small 1 large | - (not used for measurement) |
| Well Plate | 4 | - |
| Paper Towel | Sufficient amount | - |
| Pipette | 1 | ±0.001 ml |
| Analytical Balance | 1 | ±0.001 g |
| Oven | 1 | - |
| Plastic Wrap | 1 | - |
| Graduated cylinder | 1 | ±0.5 ml |
| Safety Goggles | 1 | - |
| Lab coat | 1 | - |

**Trial Run**

1. Ensures the selected pH range (3–7) is appropriate for mung bean seeds to germinate
2. Test the citric acid dilution method to determine the ratio of citric acid to water, as well as how much citric acid solution should be added each time (the increment) to allow precise adjustment.
3. Test the temperature and duration for putting the germinated beans in the oven to ensure proper dryness

## Procedure:

**I. Preparation of Citric Acid Solutions**

1. Weigh 5 grams of citric acid using an electronic balance and dissolve it in 200 mL of distilled water to make the acid solution.
2. Measure 50 mL of distilled water with a graduated cylinder. Pour 50ml in every beaker for 5 beakers; label them pH 3, pH 4, pH 5, pH 6, and pH 7.
3. Calibrate the pH probe using buffer solution with pH 4 and 7.
4. Place the pH probe in the beaker and gradually add 0.15 mL of the citric acid solution at a time while stirring. Decrease the increment of addition when the pH approaches the desired pH
5. Allows to probe to stay in the solution for 2 minutes to ensure the pH remains stable
6. Repeat the step 4-5 for all five beakers with different pH (pH 3.00, 4.00, 5.00, 6.00, or 7.00 ±0.01).

**II. Preparation of Bean Samples**

7. Measure and record the initial mass of every two Vigna radiata beans as one sample using an analytical balance. Repeat for all samples.
8. Cut paper towels into 1.5 cm x1.5 cm squares and place them in each well.
9. Place two beans in each well

**III. Watering and Incubation**

10. Add 1 mL of pH solution (or distilled water for the control group) to each well using a pipette, ensuring the beans are submerged.
11. Seal the entire well plate with plastic wrap to prevent evaporation.
12. Wrap the well plate with 3 layers of paper towel to block light.
13. Place the well plate in an area with minimal exposure to any source of light.
14. Monitor temperature throughout the 5-day incubation period to ensure stable environmental conditions.
15. Check the seeds daily for qualitative observation

**IV. Post-Germination Measurements**

16. After 5 days, take out the beans from the well plate.
17. Remove the water on the surface using paper towel
18. Dry the beans in an oven at 80°C for 30 minutes to remove moisture content. Put a thermometer in the oven to keep track of the temperature
19. Measure and record the final mass of the two beans in each sample.

Safety, Environmental and Ethical consideration:

**Citric Acid:**

- Risk: Skin and eye irritation.
- Precaution: Gloves, lab coat and goggles should always be worn to avoid direct contact. If contact occurs, the infected part should be immediately rinsed with running water for 10-15 minutes.
- Glass containers should be handled carefully to prevent breakage, which could lead to accidental contact with acid.
- Citric acid solution should be diluted before disposal to prevent water contamination

**Oven (80°C):**

- Risk: Burns or fire hazards.
- Precaution: Use heat-resistant gloves

**Ethical**:

- No animals/humans involved. Using plant seeds causes no harm to living organisms, aligning with ethical research standards.

## Data analysis:

Raw data:

**Qualitative observation:**

- Seeds treated with pH solution of 6, 7 germinated more quickly (within 24 hours)
- The control group (distilled water) demonstrated the most growth, with roots appearing white and healthy.
- Seeds treated with citric acid solution all appeared slightly brownish compared to the control

*Figure 1.* Picture of the mung beans after 5 days ( Top two rows are control; Bottom two rows are pH3 samples)

**Quantitative data:**

**Table 4. Initial and Final Biomass (in grams) of mung beans treated with pH3 solution**

| Sample | Initial Mass (±0.001 g ) | Mass after 5 days (±0.001 g ) |
|---|---|---|
| 1 | 0.119 | 0.150 |
| 2 | 0.142 | 0.151 |
| 3 | 0.142 | 0.153 |
| 4 | 0.105 | 0.135 |
| 5 | 0.139 | 0.160 |
| 6 | 0.127 | 0.130 |
| 7 | 0.123 | 0.130 |
| 8 | 0.140 | 0.138 |
| 9 | 0.130 | 0.148 |
| 10 | 0.156 | 0.156 |
| 11 | 0.143 | 0.114 |
| 12 | 0.140 | 0.156 |

The tables of initial and final biomass of mung beans treated with pH 4-7 solutions are included in the appendix.

Uncertainty of raw measurement:
The uncertainty of raw data is determined by precision of the analytical balance with uncertainty of ±0.001 g. Given the data generally ranges from 0.1-0.2, the uncertainty is 0.5% to 1% of the measured values. This small percentage uncertainty is negligible compared to the overall measured biomass.

**Data processing:**

To demonstrate the impact of pH on the germination, the change in biomass, which reflects the extent and success of germination is calculated.

**Formula 1. Calculation of the change in biomass**

Change in biomass = Final Biomass - Initial Biomass

Change in biomass for sample 1 of pH 3 = 0.150-0.119
= 0.031g

Outliers:

By eliminating outliers, the accuracy and reliability of the result will be improved as the trend can better represent the majority of the samples.

The IQR method is used to remove outliers.

1. Find Q1 and Q3
    - Using trials of pH 3 as an example: Q1=0.0015 Q3=0.0195
2. Find IQR by Q3 - Q1
    - IQR= 0.0195 - 0.0015 =0.018
3. Determine the Lower and Upper Bound:
    - Lower bound = Q1 - 1.5 × IQR = 0.0015 -1.5 ×0.018 = -0.0255
    - Upper bound = Q3 + 1.5 × IQR = 0.0195 +1.5 ×0.018= 0.0465
4. Any data point below the lower bound or above the upper bound is considered an outlier.
    - -0.029 is the outlier among pH3 samples

**Table 5. Change in biomass (g) for all samples (pH 3-7 and control) with outliers labelled**

| Sample | pH3 | pH4 | pH5 | pH6 | pH7 | Control |
|---|---|---|---|---|---|---|
| 1 | 0.031 | 0.004 | 0.026 | 0.052 | 0.011 | 0.007 |
| 2 | 0.009 | 0.015 | 0.011 | -0.005 | 0.014 | 0.03 |
| 3 | 0.011 | 0.011 | 0.057 | 0.045 | 0.028 | 0.065 |
| 4 | 0.03 | 0.014 | 0.093 | **0.093(outlier)** | 0.049 | 0.015 |
| 5 | 0.021 | 0.022 | 0.037 | 0.021 | 0.047 | 0.022 |
| 6 | 0.003 | -0.008 | 0.054 | 0.033 | 0.026 | 0.023 |
| 7 | 0.007 | 0.009 | 0.011 | 0.016 | 0.055 | 0.014 |
| 8 | -0.002 | -0.02 | 0.016 | 0.033 | 0.073 | 0.016 |
| 9 | 0.018 | -0.012 | 0.014 | 0.048 | 0.053 | 0.14 |
| 10 | 0.000 | -0.017 | 0.027 | 0.027 | 0.038 | **0.0015(outlier)** |
| 11 | **-0.029(outlier)** | 0.034 | 0.062 | 0.028 | 0.027 | 0.008 |
| 12 | 0.016 | 0.01 | 0.041 | 0.020 | 0.026 | 0.032 |

<u>Mean calculation:</u>

The mean of the change in mass for every pH is calculated to allow the comparison of the overall effects of different pH treatments.

**Formula 2. Calculation of Mean**

$$\text{Mean } (\overline{x}) = \frac{\sum x}{n}$$

$\sum x$ = sum of the mass of all samples for a given pH

n = number of samples

**Table 6. Mean Change in biomass (g) per pH condition**

| Ph | Mean change in biomass (g) |
|---|---|
| 3 | 0.013 |
| 4 | 0.005 |
| 5 | 0.037 |
| 6 | 0.028 |
| 7 | 0.037 |
| Control | 0.035 |

Standard deviation:

Standard deviation, measuring the variability and spread of data, helps us understand the consistency of the data and indicates reliability and precision of the result.

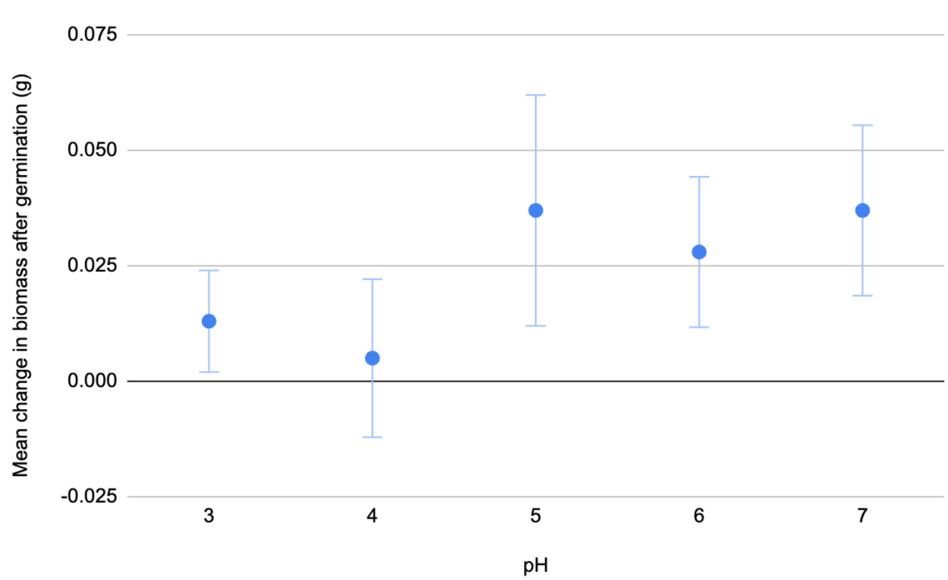
**Formula 3. Standard Deviation**

$$\sigma = \sqrt{\frac{\sum (Change\ in\ biomass - mean\ change\ in\ biomass)^2}{number\ of\ sample\ per\ pH}}$$

**Table 7. Standard Deviation per pH condition**

| pH | Standard Deviation |
|---|---|
| 3 | 0.011 |
| 4 | 0.017 |
| 5 | 0.025 |
| 6 | 0.016 |
| 7 | 0.018 |
| Control | 0.038 |

**Graph 1. Mean change in biomass (g) vs pH of solution with error bar representing standard deviation (obtained by Google sheet software )**



ANOVA Test:

The large overlapping error bars suggest the data points may not be significantly different. Therefore, ANOVA test is performed to determine if the means are statistically different by comparing means of multiple groups .

| Source of Variation | SS | MS | F | P-value | F crit |
|---|---|---|---|---|---|
| Between Groups | 0.01008 | 0.002521 | 7.5809 | 0.00006608 | 2.5463 |
| Within Groups | 0.01762 | 0.0003325 | | | |
| Total | 0.0277 | | 0.0004861 | | |

*Table 8.* Result of ANOVA Test using Excel software

The test statistic **F = 7.5809** exceeds the critical value of **2.5463**; **p-value of 0.00006608** is significantly lower than **0.05**. Therefore, the null hypothesis that all pH conditions result in the same biomass can be rejected. This indicates that the biomass of mung beans treated with different pH have a statistical significance and is not only caused merely by experimental variability or chance.

## Analysis and Conclusion:

Although the result of the ANOVA test indicates a statistically significant difference in biomass of different pH levels, Graph 1. Suggests there is no clear trend and these differences do not follow a consistent pattern. This means that pH may have some effect on the germination, but no correlation can be concluded. Therefore, my study does not support my initial hypothesis, which stated that lowering the pH would lead to a decrease in biomass. Instead, the irregular distribution of data points and overlapping error bars suggests that other factors, such as random error or flaws in experimental design, may have impacted the result .

Previous study has shown that the seeds of *Vigna radiata* germinate best in near-neutral pH, with extreme acidic or basic conditions inhibiting enzyme activities and germination(Doucoure). As my study demonstrates no clear trend, it shows no alignment with previous findings. In conclusion, the data do not support the hypothesis, nor agree with previous studies. Thus, further investigation or refinement of the experimental method needs to be done to reduce random error and improve precision.

## Evaluation:

<u>Strength of this study:</u>

- Preliminary trial: Conducting trial run allows for the testing of the practicality and effectiveness of the methodology. Adjustments in the methodology have been made to improve the accuracy of pH value of solution and the consistency of the growing environment of seeds.
- Precise measurement tools: The use of analytical balance and pipette (both with an uncertainty of ±0.001) ensures the precision and accuracy of the raw data and minimizes error caused by measurements.

<u>Weakness of the study:</u>

1. Quality of seed:

   <u>Explanation:</u> The quality of seed refers to the characteristics that influence the ability for seeds to germinate and grow. These characteristics include viability, vigor and health. In my study, the quality of the seed is controlled from two aspects: type (same brand and batch) and size (relatively uniform sizes). However these two factors alone are not enough to provide insight into the quality of seed.

   <u>Improvement:</u> Through research, it is found that storing conditions, including humidity, duration and temperature has a great impact on the quality of seed (Gebeyehu). Therefore, the quality can be better ensured by sourcing directly from farms or choosing reliable brands that store seed in optimal condition. Moreover, increasing the sample size could also reduce the impact of varying seed quality on the precision of data.

2. Number of seeds per sample:

   <u>Explanation:</u> In this experiment, each sample was made up of only two seeds. This small value not only challenges the precision of the balance when measuring mass, it also means that the result is significantly affected by individual seed variability. The impact of this weakness is clearly demonstrated in graph 1 with large error bars, which significantly decrease the reliability of the result

   <u>Improvement:</u> Each sample should consist of more than two seeds—ideally, at least five or ten seeds. This will make the result less affected by seed variability and limited precision of balance, providing a more precise and trustworthy result. In addition, a larger container such as a beaker needs to replace the well plate to ensure sufficient space for seeds to germinate and grow.

## Suggestions for further study:

1. Short duration of the experiment (5 days):

   Explanation: The experiment only lasts for 5 days because it is designed to examine the germination process. Successful germination does not guarantee the long term health of plants as it fails to acknowledge the impact of acid rain on the long term growth.

   Improvement: Extending the experiment longer to around 14 days allows us to collect data on the impact of acidic solution on seedling growth. This data can be used to reflect the long-term pH on plant growth and biomass accumulation.

2. Use of Hydroponic Growing Method:

   Explanation: The experiment uses hydroponic growing methods. However, plants typically grow in soil in the natural world. Soil contains many substances and introduces variables (nutrient content, microbial interactions, soil structure) that affect the growth of plants, and they are ignored in this study. Thus, the result may not fully represent how plants would behave in the natural soil environment.

   Improvement: A follow-up experiment could be done by growing mung beans in soil and comparing the result with this study. This comparison would improve the relevance of the study's findings to real-world agricultural and environmental protection practices.

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# Appendix:

**Germination result records:**

*Figure 2.* Picture of the mung beans after 5 days ( Top two rows are pH6; Bottom two rows are pH7)

*Figure 3.* Picture of the mung beans after 5 days ( Top two rows are pH4; Bottom two rows are pH5)

**Table 9. Initial and Final Biomass (in grams) of Mung beans treated with pH4 solution**

| Sample | Initial Mass (±0.001 g ) | Mass after 5 days (±0.001 g ) |
|---|---|---|
| 1 | 0.136 | 0.140 |
| 2 | 0.154 | 0.169 |
| 3 | 0.145 | 0.156 |
| 4 | 0.159 | 0.173 |
| 5 | 0.166 | 0.188 |
| 6 | 0.135 | 0.127 |
| 7 | 0.157 | 0.166 |
| 8 | 0.121 | 0.101 |
| 9 | 0.153 | 0.141 |
| 10 | 0.139 | 0.122 |
| 11 | 0.175 | 0.209 |
| 12 | 0.127 | 0.137 |

**Table 10. Initial and Final Biomass (in grams) of Mung beans treated with pH5 solution**

| Sample | Initial Mass (±0.001 g ) | Mass after 5 days (±0.001 g ) |
|---|---|---|
| 1 | 0.151 | 0.177 |
| 2 | 0.151 | 0.162 |
| 3 | 0.156 | 0.213 |
| 4 | 0.141 | 0.234 |
| 5 | 0.136 | 0.173 |
| 6 | 0.151 | 0.205 |
| 7 | 0.118 | 0.129 |
| 8 | 0.110 | 0.126 |
| 9 | 0.142 | 0.156 |
| 10 | 0.147 | 0.174 |
| 11 | 0.141 | 0.203 |
| 12 | 0.138 | 0.179 |

**Table 11. Initial and Final Biomass (in grams) of Mung beans treated with pH6 solution**

| Sample | Initial Mass (±0.001 g ) | Mass after 5 days (±0.001 g ) |
|---|---|---|
| 1 | 0.171 | 0.223 |
| 2 | 0.150 | 0.145 |
| 3 | 0.153 | 0.198 |
| 4 | 0.162 | 0.255 |
| 5 | 0.151 | 0.172 |
| 6 | 0.141 | 0.174 |
| 7 | 0.115 | 0.131 |
| 8 | 0.143 | 0.176 |
| 9 | 0.153 | 0.201 |
| 10 | 0.127 | 0.154 |
| 11 | 0.142 | 0.170 |
| 12 | 0.113 | 0.133 |

**Table 12. Initial and Final Biomass (in grams) of Mung beans treated with pH7 solution**

| Sample | Initial Mass (±0.001 g ) | Mass after 5 days (±0.001 g ) |
|---|---|---|
| 1 | 0.135 | 0.146 |
| 2 | 0.145 | 0.159 |
| 3 | 0.150 | 0.178 |
| 4 | 0.141 | 0.190 |
| 5 | 0.180 | 0.227 |
| 6 | 0.178 | 0.204 |
| 7 | 0.172 | 0.227 |
| 8 | 0.150 | 0.223 |
| 9 | 0.136 | 0.188 |
| 10 | 0.152 | 0.190 |
| 11 | 0.148 | 0.175 |
| 12 | 0.137 | 0.163 |

**Table 13. Initial and Final Biomass (in grams) of Mung beans treated with distilled water**

| Sample | Initial Mass (±0.001 g ) | Mass after 5 days (±0.001 g ) |
|---|---|---|
| 1 | 0.115 | 0.122 |
| 2 | 0.137 | 0.167 |
| 3 | 0.146 | 0.211 |
| 4 | 0.143 | 0.158 |
| 5 | 0.172 | 0.194 |
| 6 | 0.187 | 0.210 |
| 7 | 0.113 | 0.137 |
| 8 | 0.122 | 0.138 |
| 9 | 0.136 | 0.276 |
| 10 | 0.167 | 0.182 |
| 11 | 0.110 | 0.118 |
| 12 | 0.166 | 0.198 |